

Airframe Design Theory Exploration: Part 2

Braxton Lopez

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Vocabulary

0.1 Non-Dimensional Numbers

The power of nondimensionalization is that it can express a complicated relationship succinctly and universally, and it is the basis for all wind tunnel testing. For example, the drag of a given airfoil is a function of many things, the angle of attack, air density, viscosity, speed of sound, freestream speed, its size, etc.:

$$D = f(\alpha, \rho, \mu, V_\infty, a, c) \quad (1)$$

However, we can express the same relationship in terms of a non-dimensional drag coefficient as:

$$C_D = f(\alpha, Re, M) \quad (2)$$

The power of this form is that the drag coefficient doesn't depend on density directly (for example), but rather on Reynolds number. Thus, if we create the same Re and M with a scaled geometry, we can determine the drag coefficient that would be experienced on the full airplane. Similarly, for a given shape, we can tabulate or calculate lift and drag coefficients as a function of α , Re and if high speeds are important M and re-use them for various applications.¹

Reynolds Number The Reynolds Number:

$$Re = \frac{\rho V c}{\mu} \quad (3)$$

where ρ and μ are the fluid's density and dynamic viscosity respectively, V is the freestream (or local) speed, and c is some relevant length scale used in normalization, which for an airfoil would be the chord length and for a wing would be the mean aerodynamic chord. The Reynolds number is a ratio of inertial to viscous forces. For a commercial transport, the Reynolds number is typically in the millions or tens of millions. In other words, outside of the boundary layer, the air speed is so high that inertial forces are much, much more important than viscous forces. For a small UAV, the Reynolds number might be of order 10,000². For aerodynamics generally Reynolds number and Mach number are the most important.³

¹A. Ning, *Flight vehicle Design*, 2018. pg. 35.

²A. Ning, *Flight vehicle Design*, 2018. pg. 34.

³A. Ning, *Flight vehicle Design*, 2018. pg. 35.

Mach Number The Mach Number:

$$M = \frac{V}{a} \quad (4)$$

is a ratio between the freestream (or local) speed and the speed of sound. In other words, a Mach number of one means that the vehicle is moving at the speed of sound. Mach numbers less than 1 are called subsonic, at 1 is called sonic, and above 1 is called supersonic. Mach numbers larger than about 5 are called hypersonic, although the number 5 is somewhat arbitrary as there is no fundamental change in flow physics right at that point like there is at the sonic point. A flow field with Mach numbers less than about 0.3 is essentially incompressible, and thus the Mach number is not a relevant parameter for that flow. For aerodynamics generally Reynolds number and Mach number are the most important.⁴

Lift and Lift Coefficients Lift is the force created by the pressure differential between the low-pressure (suction) surface and high-pressure (pressure) surface of the wing. This pressure differential is created by the flow of air around the shape of the airfoil, and this lifts the wing upwards. Lift is always defined as being perpendicular to the freestream direction. As the airstream passes by the plane, the airfoil separates the stream of air, creating a low-pressure zone above the wing and a low-pressure zone below it. The high pressure zone pushes the wing upwards, and as the airfoil's camber increases, this pressure difference grows, creating more lift. An increase in the airstream's flow speed will also generate an increased amount of lift; however, its important to note (as seen in figure 12) that the amount of lift produced depends greatly on the angle of attack α .

The lift coefficient of an airfoil is related to the area of the airfoil by the equation:

$$c_l = \int_0^1 (c_{p_l} - c_{p_u}) d\frac{x}{c} \quad (5)$$

where c_{p_l} and c_{p_u} are the pressure coefficient on the lower and upper surface respective, and $\frac{x}{c}$ is the normalized axial position along the airfoil from 0 to 1⁵. Optimized designs will be able to produce lift efficiently with minimal drag. If the wing area is fixed then increasing span reduces maximum lift coefficients. The maximum lift coefficient that is achievable before stall is denoted as $c_{l_{max}}$ in 2D or $C_{L_{max}}$ in 3D.

Drag and Drag Coefficients Drag is the aerodynamic force that opposes an aircraft's motion through the air. It acts opposite the direction of motion and is generated by the interaction between a solid body and the surrounding fluid (typically air). Its always defined as being parallel to the freestream direction (perpendicular to lift).

Because an airfoil is 2D, the drag is actually drag per unit span, represented by D' . While in 2D, the coefficient of drag is defined by:

$$c_d = \frac{D'}{q_\infty c} \quad (6)$$

where q_∞ is the dynamic pressure. A three-dimensional body, like a wing, is defined similarly but with an extra length dimension in the denominator:

$$c_d = \frac{D}{q_\infty S_{ref}} \quad (7)$$

⁴A. Ning, *Flight vehicle Design*, 2018. pg. 35.

⁵A. Ning, *Flight vehicle Design*, 2018. pg. 40.

The type of reference area S_{ref} used must be specified when providing coefficients. For more information on drag, see Part 1 of this document.

The drag coefficient expresses the ratio between the drag force on an object and the dynamic pressure acting over a reference area. Its a dimensionless number that tells you how much aerodynamic resistance an object experiences as it moves through a fluid such as air or water.

The variation in the drag coefficient is called the drag polar (For more information, see Part 1 of this document). For low Mach numbers, the total drag coefficient is given by the equation:

$$C_D = C_{D_P} + \frac{C_L^2}{\pi A Re} \quad (8)$$

where C_{D_P} is the coefficient of parasitic drag. The drag of a given airfoil is a function of many things, the angle of attack, air density, viscosity, speed of sound, freestream speed, its size, etc.:

$$D = f(\alpha, \rho, \mu, V_\infty, a, c) \quad (9)$$

However, we can express the same relationship in terms of a non-dimensional drag coefficient as:

$$C_D = f(\alpha, Re, M) \quad (10)$$

The power of this form is that the drag coefficient doesn't depend on density directly (for example), but rather on Reynolds number. Thus, if we create the same Re and M with a scaled geometry, we can determine the drag coefficient that would be experienced on the full airplane. Similarly, for a given shape, we can tabulate or calculate lift and drag coefficients as a function of α , Re and if high speeds are important M and re-use them for various applications.⁶

For cases in which you might only operate the plane within a narrow range of Re and M and where performance does not vary significantly with these parameters, the lift, drag, and moment coefficients are effectively only a function of angle of attack.⁷

Moment Coefficients The moment coefficient is a dimensionless aerodynamic coefficient that describes the aerodynamic moment (torque) acting on a body—usually an airfoil, wing, or entire aircraft—about a chosen reference point. It is the sum of all 3 dimensional moments.

Moment coefficients for a given airfoil are a function of the angle of attack, Reynolds number, and Mach number. For cases in which you might only operate the plane within a narrow range of Re and M and where performance does not vary significantly with these parameters, the lift, drag, and moment coefficients are effectively only a function of angle of attack.⁸

Denoted by c_m in 2D and C_m in 3D, the moment coefficient is given by the equations:

$$c_m = \frac{M'}{q_\infty c^2} \quad (11)$$

$$C_m = \frac{M}{q_\infty S_{ref} c} \quad (12)$$

⁶A. Ning, *Flight vehicle Design*, 2018. pg. 35.

⁷A. Ning, *Flight vehicle Design*, 2018. pg. 41.

⁸A. Ning, *Flight vehicle Design*, 2018. pg. 35.

where c is the chord, and the type of reference area used must be specified when providing coefficients. For a wing it is typically the planform area, whereas for a fuselage the cross-sectional area is typical. For a full aircraft the trapezoidal reference area is typically used. This area is a trapezoid that matches the main portion of the wing and extends all the way to the centerline. Its not the exact same shape as the wing, but is easier to use. Any area can be used as long as it is consistent and is defined, it merely provides the length scales used in the normalization. Prior to stall, moment coefficient also has (theoretically) linear behavior and in practice is usually close to linear.

0.2 Airframe Performance

Lift Distributions A lift distribution is generally a graph that describes how the lift force is spread out along the wingspan of a finite wing, from the root to the tip. It's a spanwise function, not a single number (See Figure 1). Chord and twist are the biggest factors in determining the shape of the lift distribution with sweep next in importance. If the wing is untwisted, then an elliptic distribution of chord can be used to create an elliptic lift distribution. Although an elliptic wing can produce an elliptic load, there is no real advantage to this shape as an elliptic distribution of lift can be created with any wing, properly twisted. Unfortunately, the lift distribution cannot be determined in isolation. The lift coefficient is related to the lift distribution through its definition:

$$c_l = \frac{L'}{q_\infty c} \quad (13)$$

Note that the lift coefficient distribution is a function of the lift distribution and the chord distribution. An ideal lift distribution for a planar wing, from an aerodynamic point of view, is elliptic. When we consider not just aerodynamics but also structures, the optimal lift distribution is generally loaded inboard further as shown in Figure 1. By shifting more of the load inboard, the wing weight can be reduced, primarily because of reduced bending loads, while only sacrificing a small amount of induced drag. For the aerostructural optimum a typical inviscid span efficiency is about 0.98 (compared to 1 for elliptic). We tradeoff a small increase in induced drag, for a decrease in weight, such that the overall system has improved.⁹.

One of the primary challenges of wing design is to create a chord/twist/sweep combination with a desirable lift distribution that balances induced drag and structural considerations, and a lift coefficient distribution that balances efficiency with preventing tip-stall. This is not always easy, and like most engineering decisions, requires making some tradeoffs¹².

Stall Stall occurs because the angle of attack becomes too large and the flow separates from the airfoil, causing a significant increase in drag and a decrease in lift¹³. When the angle of attack is too large, the airstream no longer follows the curvature of the airfoil and will instead create a low-pressure zone behind the wing, generating large amounts of drag. The maximum lift coefficient that can be achieved before stall is denoted by $c_{l_{max}}$ or $C_{L_{max}}$ in 3D. In figure

⁹A. Ning, *Flight vehicle Design*, 2018. pg. 78.

¹¹A. Ning, *Flight Vehicle Design*, 2018, pg. 79.

¹²A. Ning, *Flight vehicle Design*, 2018. pg. 78.

¹³A. Ning, *Flight vehicle Design*, 2018. pg. 41-42.

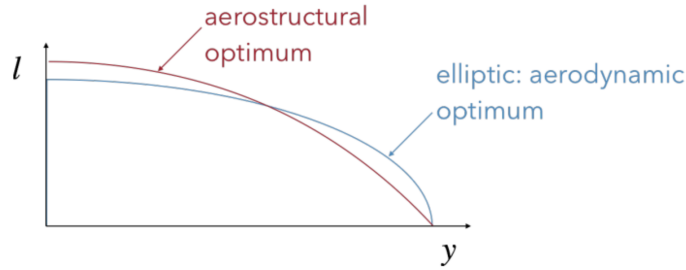


Figure 1: The elliptic distribution in blue is the aerodynamic optimum. The aerostructural optimum is loaded further inboard to balance tradeoffs in induced drag and wing weight.¹¹

2, you can see this by observing where $c_{l_{max}}$ is achieved, and the coefficient of lift begins to decrease. For a 2D airfoil, the lift coefficient prior to stall can be computed as:

$$c_l = m(\alpha - \alpha_0) \quad (14)$$

where m is the lift curve slope. Critical wing theory is also used to determine when stall might occur. A wing should optimally stall at the roots before it stalls at the tips, as this increases stability, and you can control this by changing the wing's twist to create washout. Optimally, a wing should stall at the roots first, as the tips are further away from the plane's fuselage, therefore creating a greater moment about the plane when they do stall, decreasing stability.

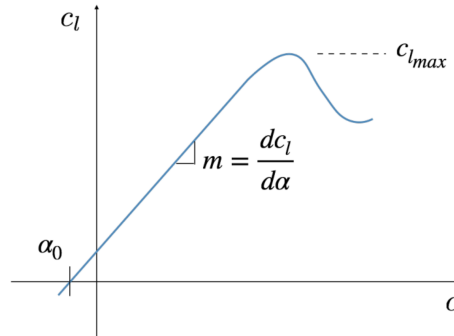


Figure 2: A notional lift curve slope highlighting the zero-lift angle of attack, lift curve slope, and maximum lift coefficient before stall.¹⁵

Stall Speed Stall speed is the speed at which at which an airfoil will begin to experience stall (See above). Any speeds less than the stall speed will result in unstable flight due to this stall. The stall speed is the minimum speed required for the wing to produce enough for the plane to be stable. You can adjust twist, sweep, and other geometric constraints in order

¹³A. Ning, *Flight Vehicle Design*, 2018, pg. 42.

to decrease the this¹⁶. The stall speed for level flight is given by:

$$V_s = \sqrt{\frac{2W}{C_{L_{max}}\rho S_{ref}}} \quad (15)$$

Stall speed is important when considering takeoff at high altitudes, where you want to maximize lift without stalling. The lower the stall speed, the more stable your aircraft will be at low speeds. This equation tells us what we can do if we want a slower stall speed, and thus a shorter runway for takeoff/landing. We can't control air density, but this equation does explain why we need a longer runway at airports at high altitudes. Decreasing stall speed is a common reason for increasing wing area, but that choice comes with other design compromises. Clearly, increasing $C_{L_{max}}$ has a major effect on decreasing stall speed, as evidenced by equation 8¹⁷.

1 Theory

Potential Flow Theory Potential flow theory is an idealization of real fluid motion in which the airflow is assumed to be inviscid, incompressible, and irrotational. Assuming these conditions, the velocity field can be described using a scalar potential that satisfies Laplace's equation, making the flow easier to compute. Potential flow often provides generally accurate predictions of pressure distributions and lift for streamlined bodies, as long as viscous effects like boundary layers and flow separation are not dominant.

The governing equation for potential flow is known as Laplace's equation:

$$\nabla^2\phi = 0 \quad (16)$$

For this function to be valid, you must assume both incompressible and irrotational flow. For aerodynamics, an incompressible flow occurs for low Mach numbers, approximately less than $M = 0.3$.

Potential flow is an extremely useful theory for low-speed aerodynamics, and forms the basis of panel methods widely used in many conceptual design studies¹⁸. Methods using potential flow theory can be useful up to low transonic mach numbers. It cannot; however, be used to find the boundary layers, viscous drag, or flow separation.

Because Laplace's equation is linear, there are various elementary solutions. There are four main solutions: one for uniform flow, one for a source, one for a vortex, and one for a doublet. For more information on this, see pages 42-43 in *Computational Aerodynamics* by Andrew Ning. In summary, potential flow theory is used to solve for aerodynamic flowfields around slow moving objects

Note 1: The definition for vorticity is:

$$\vec{\omega} = \nabla \times \vec{V} \quad (17)$$

Note 2: A flow field is defined as being irrotational if:

$$\nabla \times \vec{V} = 0 \quad (18)$$

¹⁶A. Ning, *Flight vehicle Design*, 2018. pg. 81.

¹⁷A. Ning, *Flight vehicle Design*, 2018. pg. 156.

¹⁸A. Ning, *Computational Aerodynamics*, 2018. pg. 41.

Vortex Filaments and Horseshoe Vortices Vortex filaments are thin, two dimensional lines. If you were to look at them from the side, there would be a single dot in the center of the vortex wake created by the airplane's travel. This point is extended parallel to the direction in which the wake (and airplane) is traveling. In other words, a vortex filament is a straight line that defines where the center of a three dimensional wake is. They are created under the assumption that the airflow is both incompressible and irrotational, implying that we are still governed by Laplace's equation (the potential flow theorem). Vortex filaments are an idealized concept in which these vortices are "concentrated" into this singular point where the line is.

The circulation of one of these vortices is defined by the right hand rule, the fingers curl around in the direction of induced velocity and the thumb points along the line defining the direction of circulation¹⁹. It's velocity has a magnitude:

$$|V| = \frac{\Gamma}{2\pi r} \quad (19)$$

A horseshoe vortex is a u-shaped (almost like a horseshoe) system of vortices, composed of three separate vortex filaments. The first one is called a bound vortex filament. This vortex filament is parallel to the span of the wing, and these vortices are generated around the wing. The other two are called trailing vortices, which are generated around the tip of the wings. If you were to look at wing from behind the plane, the bound vortex filament would be a solid line parallel to the wing's span, and the trailing vortex filaments would look like two dots, one on each wing's tip. In the figure below, we see these, as well as a starting vortex filament. The starting filament is typically negligible, and we simply assume the trailing vortices are infinitely long (its typically assumed that the starting filament isn't there). The bound vortices generate lift, and the trailing vortices generate induced drag.

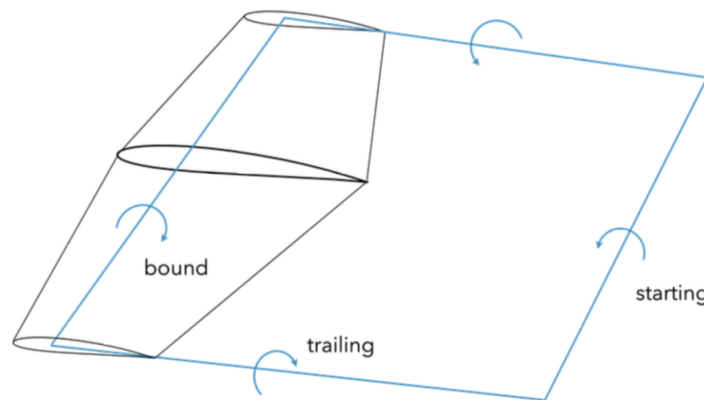


Figure 3: Closed path vortex filaments produced by a wing.²⁰

¹⁹A. Ning, *Computational Aerodynamics*, 2018. pg. 120.

²⁰A. Ning, *Computational Aerodynamics*, 2022, pg. 123.

Using Horseshoe Vortices to Model Lifting Surfaces As mentioned above (See *Vortex Filaments and Horseshoe Vortices*), vortex filaments are generated assuming incompressible and irrotational airflow. This means that we are still governed by Laplace's equation and the potential flow theorem. As mentioned above, lift is also generated by the bounding vortices, and induced drag is caused by the trailing vortices.

If you wanted to use horseshoe vortices to model a lifting surface, you would first want to determine which solver you will use. The method explained below will be the VortexLattice method.

The VortexLattice method (VLM) uses the thin air foil assumption so that lifting surfaces are flat (not curved), and changes in twist are handled in the boundary condition and not the geometry. A wing may be represented with multiple flat surfaces (several panels, both chordwise and spanwise), for example, a wing and winglets. Curved surfaces require a more general 3D panel method; however, in VLM you can use horseshoe vortices to model the lift. In this method, a lifting surface is divided into panels. Assign each panel a horseshoe vortex as shown in Figure 4. All three vortex filaments (two trailing vortices and one bounding vortex) will have the same circulation Γ ²¹. The trailing vortices extend into the +x direction of the body axes²².

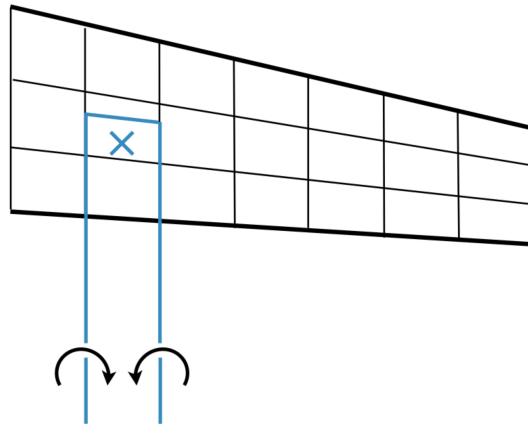


Figure 4: An example paneling on a lifting surface. A horseshoe vortex is depicted on one panel, and each panel will have its own horseshoe vortex. The x denotes the control point for the given panel.²⁴

A VLM can have both spanwise and chordwise panels. If only spanwise panels are used, it is called a Weissinger formulation. This simplification is appropriate for high aspect ratio wings where resolving the chord wise pressure distribution is less significant in obtaining an accurate lift distribution. Because the vortex filaments automatically satisfy the governing equations, the only remaining conditions to satisfy are flow tangency and the Kutta condition. The flow tangency condition (or no-flow- through condition) will be applied at select control points. Because each horseshoe vortex has a constant (as yet unknown) strength, we can only have one control point per panel, thus maintaining an equal number of unknowns and

²¹A. Ning, *Computational Aerodynamics*, 2018. pg. 135.

²²A. Ning, *Computational Aerodynamics*, 2018. pg. 135.

²⁴A. Ning, *Computational Aerodynamics*, 2022, pg. 135.

equations. The control point was denoted as an x in Figure 4²⁵.

A vortex lattice method uses what is called a lumped vortex method, this means that distributed vorticity along each panel is all lumped into one vortex as shown in Fig. 2.23. Even with multiple chordwise panels, this approach is still used, just with the vorticity lumped into separate vortices for each chordwise panel. Also note that in using the lift result from thin airfoil theory we have implicitly imposed the Kutta condition already. Thus, all that remains is satisfying flow tangency at the control points.

External Drag Models for Determining Stall Its necessary to use an external drag model for determining when stall might occur. Because the methods described above (including VLM) assume incompressible, irrotational and inviscid airflow, there are discrepancies once you reach a large angle of attack; these methods are only accurate until approximately $\alpha = 15^\circ$, at which point VLM can't predict airflow correctly. After this point, the boundary layers begin to separate and stall begins to occur, causing a large drop in the amount of lift being produced (and a large increase in drag). You can see this below in figures 5 and 6. Figure 5 models how the lift and drag curves vary with α (values for this example provided in part 1 of this document) and Figure 6 models how lift should vary with viscous airflow. As we can see, there is a large difference, and note that figure 6's lift curve begins to drop around $\alpha = 15^\circ$. As you can see in figure 7, drag increases as lift increases. These two values are interconnected, and after $\alpha = 15^\circ$, always use an external drag model. In the case of VortexLattice, it cannot predict many different types of drag (such as skin friction drag, wave drag, etc). It estimates these; however, they are slightly skewed from more accurate methods.

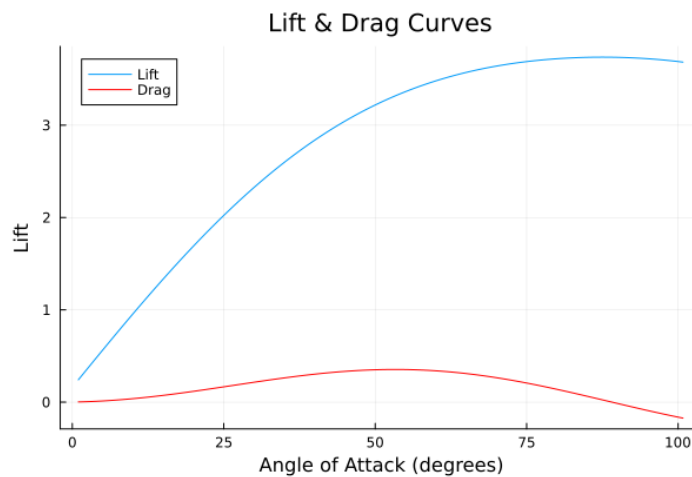


Figure 5: A description of how lift and drag vary in regards to the angle of attack. The geometric constraints for this example are described in part 1 of this document.

²⁵A. Ning, *Computational Aerodynamics*, 2018. pg. 136.

²⁶A. Ning, *Computational Aerodynamics*, 2022, pg. 82.

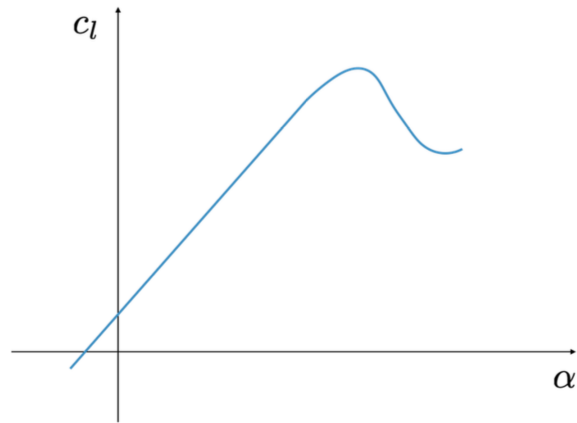


Figure 6: Viscosity causes a reduction in lift curve slope and flow separation causes stall.²⁶

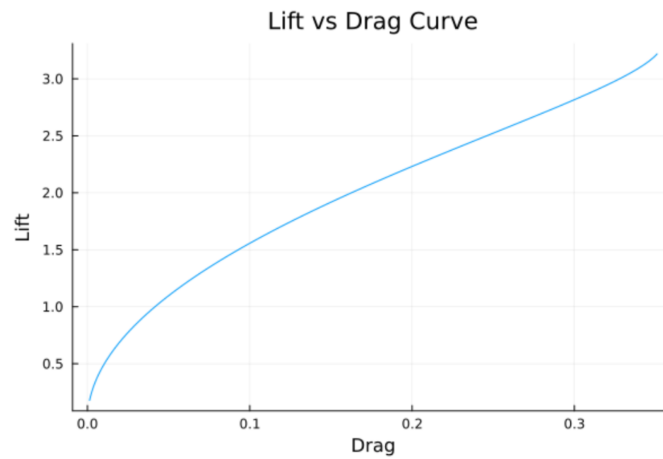


Figure 16: A description of how lift varies with regards to drag.

Figure 7: A description of how lift varies with regards to drag. The geometric constraints for this example are described in part 1 of this document.

2 Exploration

Lift Distributions Below are the results of my exploration.

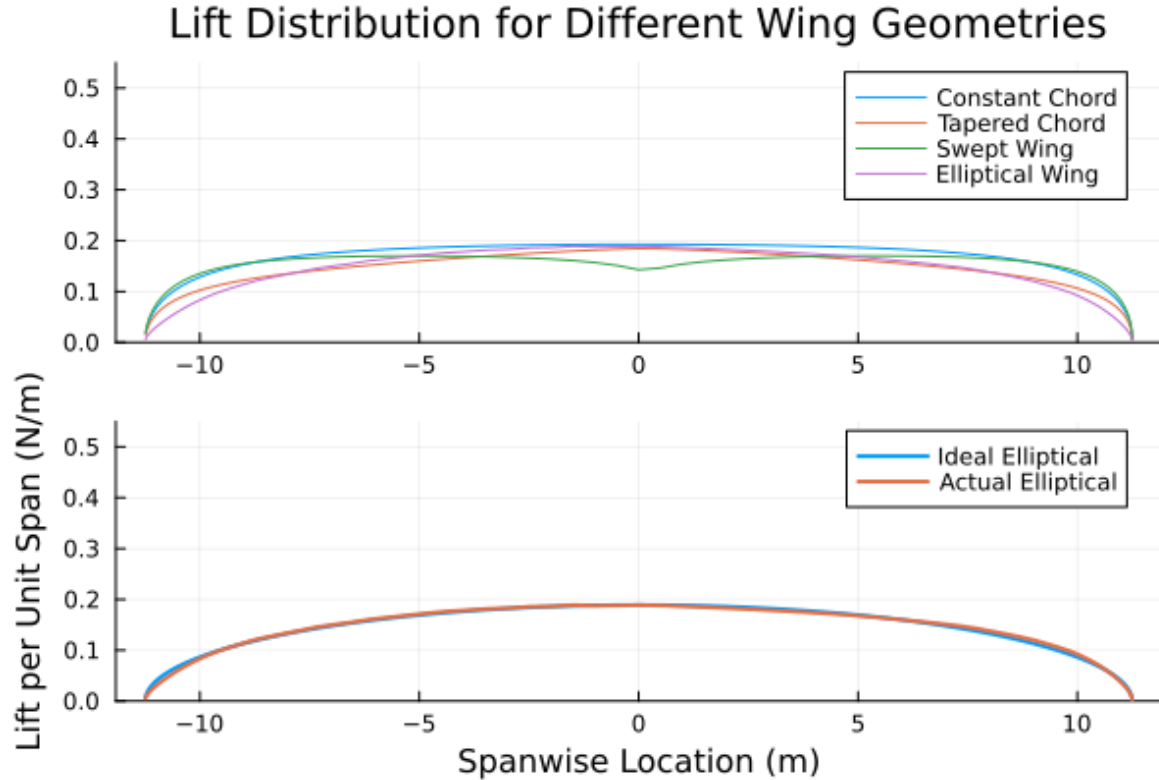


Figure 8: The top graph illustrates the lift distribution curves of four separate wing geometries which were all generated using Vortex Lattice. The bottom figure illustrates the lift distribution curve of the elliptical wing represented in the top figure, compared to what an ideal distribution for a wing should be. Please note that these curves are for an entire wing and not just one half of a wing

As you can see in the bottom graph, the ideal elliptical curve is almost identical to the actual elliptical curve generated through the Vortex Lattice method (VLM). Although it doesn't match perfectly, this illustrates that the VLM provides a fairly accurate estimation of how the wing geometry affects lift. The top graph in Figure 8 illustrates exactly how geometry affects the lift distribution curve. In the Figure above, all wings were defined as having no twist and the same wing area ($33m^2$).

When determining the total efficiency of these wing geometries, efficiency is defined by the equation:

$$e = \frac{C_L^2}{\pi ARC_{D_i}} \quad (20)$$

I found the efficiencies of the four wings (in order of most efficient to least efficient) to be:

$$e_{elliptical} = 1.012272$$

$$e_{tapered} = 0.977119$$

$$e_{constant} = 0.96783$$

$$e_{swept} = 0.89694187$$

This shows us that an elliptical wing design is the most efficient, with a tapered wing being a close second. Note that this specific value (for the tapered wing) is relative to the amount of taper and will vary. It tends to increase as the tip chord decreases. Also note that the elliptical design has an efficiency of more than 1. This can happen in the event of a non planar lift distribution (e.g. adding winglets)²⁷. In the VLM you cannot do this; however, small numerical approximations allowed this to happen.

Elliptical wing design are the most efficient due to how their geometry affects the strength of the vortices generated in their wake (these vortices create induced drag). The amount of lift being produced is strongest at the wing's roots and weakest at its tips (at the tips it should be zero). This allows for the wingtip vortices to be minimized and for downwash to be spread evenly across the wing's span. In reality there are constraints about how practical this planform can be; however, elliptical planforms are the most efficient out of these four planforms.

²⁷A. Ning, *Computational Aerodynamics*, 2022, pg. 135.